STANDARD OPERATING PROCEDURE (SOP)

JAL-SHODHAK FIELD DEPLOYMENT PLAYBOOK

Complete Step-by-Step Guide: From Pre-Float Commissioning to National Irrigation Scaled Operations

1 Phase 1: Pre-Deployment & Site Reconnaissance

A. Canal Bathymetry & Weed Density Survey

Fly a low-altitude multispectral mini-drone or download Sentinel-2 NDVI satellite imagery to identify the densest weed choke-points and locate submerged debris. Confirm canal width (>3m) and minimum water depth (>0.35m).

B. RTK Base Station & GCS Setup

Mount the RTK-GNSS tripod on a solid canal embankment point. Power on the LoRa differential telemetry transmitter (868/915 MHz). Establish centimeter-accurate coordinate correction lock with the drone's dual rover antennas.

2 Phase 2: Hardware Pre-Float Checklist

Check Item	Inspection Procedure	Expected Threshold	Action if Non-Compliant
Hull Watertightness	Inspect catamaran pontoon hatches, O-rings, and cable glands	Zero moisture inside bilge compartments	Replace silicone gasket & vacuum check
LiFePO4 Battery Pack	Check BMS telemetry on smart display	Voltage > 51.2V (State of Charge > 90%)	Connect solar fast-charger or hot-swap pack
Cutter Bar & Shredder	Verify blade clearance and lubricate tungsten drive eccentrics	Free rotation; cutter gap < 0.4 mm	Re-shim blade guide and tighten drive clamp
Screw Press & Die	Inspect slotted wedge-wire screen for residual dried fibers	Clear extrusion channels; pre-heat jacket to 60°C	Run manual cleanout cycle using water jet
Emergency Kill-Switch	Test physical latch switch and remote 433MHz RF beacon	Immediate thruster and motor cutoff (<100ms)	Do not launch until relay safety is verified

3 Phase 3: Software Stack & Autopilot Boot Procedure

Terminal Boot Sequence on NVIDIA Jetson Orin Nano:

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1. $ sudo systemctl start jalshodhak-marine-core.service   # Launches ROS2 Iron core and CAN-Bus driver
2. $ ros2 launch perception stereo_lidar_fusion.launch.py      # Initializes IP68 cameras & solid-state LiDAR
3. $ ros2 run weed_detector yolov8_waterweed_tensorrt          # Loads 40 TOPS accelerated weed segmentation
4. $ mavproxy.py --master=/dev/ttyTHS1 --baudrate 921600        # Opens bridge to STM32 ArduPilot autopilot
```

Autonomous Mission Upload (QGroundControl)

Define the canal clearing polygon. Set the survey pattern with 1.4m lane overlap. Set forward speed to 1.8 knots (0.9 m/s) and submerge the cutter table to 0.7m. Upload waypoints over encrypted MAVLink.

Canal Water Entry & Auto-Dock Setup

Back the custom low-bed trailer into the canal slipway. Release the electric winch strap. Drone floats off autonomously, engages reverse thrusters, and holds station via RTK dynamic positioning.

4 Phase 4: Autonomous Harvesting & Biofuel Offloading Cycle

Step A: In-Situ Cutting & Shredding

The drone engages the submerged reciprocating blades, severing weed roots. The intake conveyor continuously sweeps the biomass into the counter-rotating shredder, mulching 120 kg of wet hyacinth per minute.

Step B: Continuous Dewatering & Pelletizing

The progressive screw press squeezes out 85% of plant water, returning clear aerated liquid to the stream. The dried fibrous cake is extruded through the heated radial die into 8mm fuel pellets and deposited into the 350kg onboard bunker.

Step C: Precision Docking & Offloading

When the internal pellet hopper reaches 90% capacity, the drone automatically navigates to the canal-bank transfer point. A laser alignment sensor mates with the mobile hopper, discharging dry pellets via an electric auger in 4 minutes.

Step D: Post-Shift Maintenance Flush

Perform a 5-minute high-pressure freshwater wash of the cutter bar and screw press. Inspect propeller Kort nozzles for entangling fishing line. Plug into the canal-side solar charging bank.

5 National Deployment Strategy: Scaling to Millions of Canal Kilometers

How an Engineering Team Can Execute This at the State & National Level:

1. The 5-km Demonstration Pilot (MoU)

Approach the **Executive Engineer (Irrigation / Water Resources Department)** or District Collectorate. Offer a zero-cost demonstration on a notoriously choked 5 km canal stretch to measure flow velocity increase and flood prevention.

2. Securing Grants & Capital

Apply under national schemes: **Startup India Seed Fund**, **DST-NIDHI**, **BIRAC Clean Energy**, or CSR grants from national infrastructure companies (NTPC, NHPC, PowerGrid, Oil PSUs) focused on water rejuvenation.

3. The Village Cooperative Model (FPOs/SHGs)

Instead of managing every drone yourself, license the fleet to local **Farmer Producer Organizations (FPOs)** and **Women's Self-Help Groups (SHGs)**. They operate the drones, keep canals clean, and sell the biofuel pellets for sustainable rural income.

4. Integration into National Water Missions

Link real-time canal telemetry (turbidity, dissolved oxygen, flow rate, GIS cleared area) into the **National Water Informatics Centre (NWIC)** dashboard via API for real-time nationwide water monitoring.

Summary Action Plan: Build scale prototype → Run bench tests in irrigation channel → Present before-and-after drone telemetry to Irrigation Department → Scale via village cooperatives and green government tenders.